Machine Learning I

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Executive Summary

- **Goals:** use a classification algorithm taught in classes to obtain the accuracy of a set of benchmark datasets
- **Outline of the Approach:** modify an already made implementation of the algorithm on existing open-source code in order to make it robust to the data characteristic in question.
- **Summary of Results:**
  - Table comparing Accuracy/F1/Precision/Recall/AUC across datasets
  - Learning curves showing stability

Selected Algorithm

We chose the CART algorithm due to its easy learning curve.
CART stands for Classification And Regression Tree which, in itself, uses a Decision Tree to determine the accuracy of a dataset.
Decision Trees are among the most popular models in ML, as they are used not only by CART, but also by other algorithms, such as ID3 (Iteractive Dichotomiser 3) or C4.5, the latter being an extension of CART.
CART uses the Gini index as its impurity measure:
Given a data set D $= \{\langle x_i, y_i \rangle\}_{i=1}^{N}$ (with N examples),
$Gini(D) = 1 - \sum_{i=1}^{c} {p_i}^2$, where p_i is the probability of class i.

Data Characteristic

Decision trees, when working with imbalanced datasets, tend to maximize node purity, leading them to favor the majority class at the expense of the minority class. This is due to the fact that decision trees emphasize the more frequent class, which can lead to high misclassification rates for the minority classes.
There are some ways to address this issue, such as:

- Cost-sensitive learning (assign higher misclassification costs to the minority class)
- Sampling techniques (balance the dataset by oversampling the minority class or undersampling the majority class)
- Suitable prediction metrics (e.g. precision, recall, f1-score)

Proposal

For this assignment, we were given the three following set of benchmark datasets:

1. Noise or Outliers
2. Class Imbalance in Binary Classification
3. Multiclass Classification

Since CART uses binary partitioning, we ended up using the set of binary classification datasets (Class Imbalance in Binary Classification). We also used this set because of the fact that CART can deal with class imbalanced data, if we apply some of the techniques mentioned before.

Emprical Study

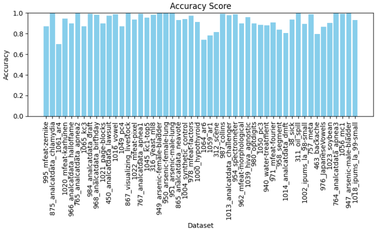
Experimental Setup:

- Datasets: binary classification datasets that suffer from class imbalance.
- Hyperparameters of the Algorithm:
  - Maximum Depth of the Tree
  - Criterion (Gini)
  - Class Weight

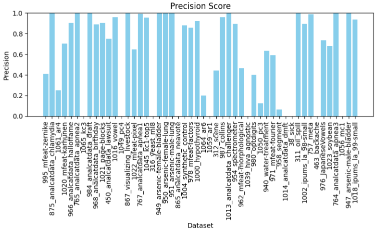
Performance Estimation Methodology:

Since most of the datasets tested for this assignment are so large, we used the holdout method, in which we split the data in two sub-sets: one for training and the other for testing.

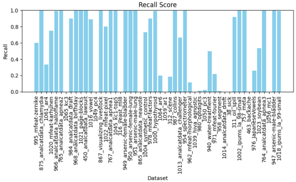
Emprical Study - Performance of the original CART algorithm



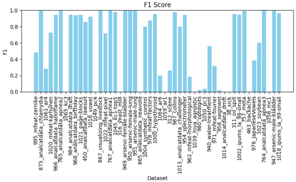
(a) Accuracy



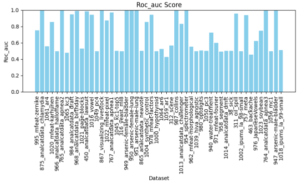
(b) Precision



(c) Recall

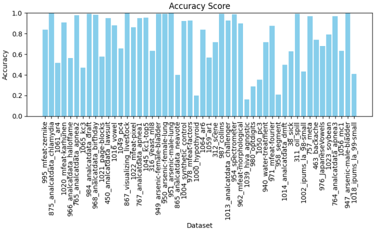


(d) F1-Score

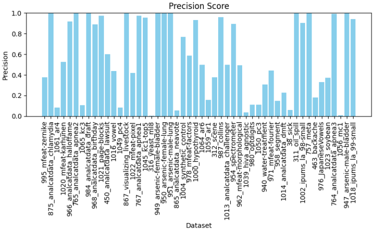


(e) ROC

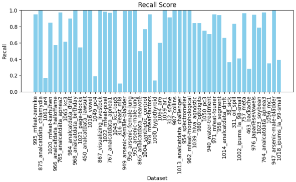
Emprical Study - Performance of the modified CART algorithm



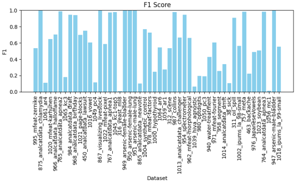
(a) Accuracy



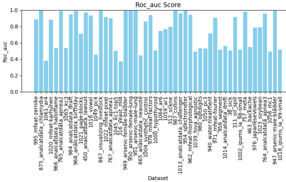
(b) Precision



(c) Recall

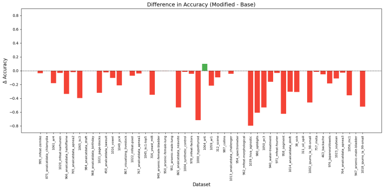


(d) F1-Score

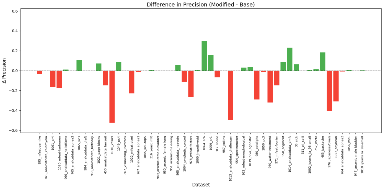


(e) ROC

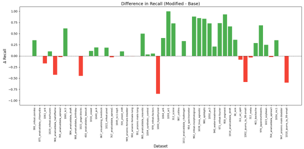
Emprical Study - Performance Comparison



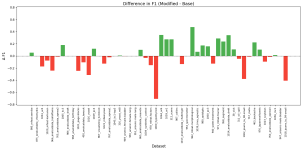
(a) Accuracy



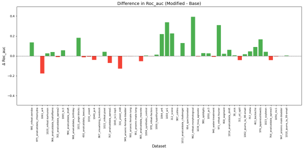
(b) Precision



(c) Recall

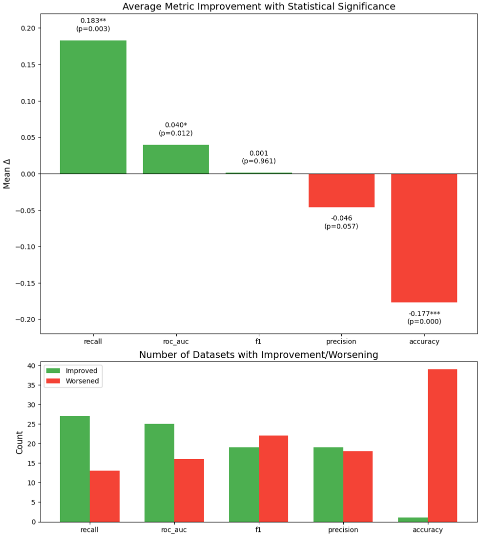


(d) F1-Score



(e) ROC

Empirical Study - Improvement Summary



Visual shorthand for p-value ranges: *** = p < 0.001 (most significant) | ** = p < 0.01 | * = p < 0.05 | (no stars) = Not statistically significant

Conclusions

References